AI Multilingual Training Video Generator

Comprehensive Technical Audit, Architecture Specification & System Assessment

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Executive Summary

The **AI Multilingual Training Video Generation Platform** is an autonomous multi-modal system designed to transform instructional educational outlines and raw curriculum text into localized video packages. The platform implements an end-to-end pipeline combining LLM script writing, rule-based duration validation, cultural localization (Arabic RTL and Bengali LTR), Edge-TTS voice synthesis, HeyGen avatar generation with local mock fallback, 1080p slide graphics via Pillow, deterministic subtitles (.SRT/.VTT), and native FFmpeg composition. This document serves as the formal engineering audit of the platform, cataloging implemented capabilities, testing metrics, and current real-world defects without modifying current application behavior.

1. System Architecture & Pipeline Breakdown

The platform coordinates 9 sequential stages via a centralized **ProjectOrchestrator** with granular state tracking in project.json manifests:

Stage	Subsystem	Key Technologies & Responsibilities
Stage 1	Script Generation	LangGraph StateGraph, GroqProvider, structured JSON extraction, deterministic duration validation.
Stage 2	Localization	LocalizationService, ar-SA (Arabic RTL), bn-BD (Bengali LTR), cultural guidance & terminology locks.
Stage 3	Speech Synthesis	EdgeTTSProvider (free WebSocket tier), exact duration measuring replacing word-count heuristics.
Stage 4	Avatar Presenter	HeyGenProvider abstraction, MockAvatarProvider local colored canvas generator.
Stage 5	Supporting Visuals	Pillow (PIL) Full HD (1920x1080) slide card renderer adapted to scene instructional roles.
Stage 6	Subtitle Synthesis	DeterministicSubtitleEngine generating drift-free .srt and .vtt cue streams.
Stage 7	Video Composition	FFmpegCompositionProvider layering presenter MP4, visual slides, audio, and burned subtitles.
Stage 8	Orchestration	ProjectManifestManager, stage idempotency, and partial pipeline resume.
Stage 9	API & Studio UI	FastAPI REST endpoints (/api/v1/projects) and Streamlit interactive dashboard.

2. Testing Status & Provider Verification Matrix

Automated Regression Suite: 162 total tests executed, **162 passed (100%)**, 0 failed, 0 errors, 0 warnings in 27.35 seconds. Passing unit and mock tests verifies internal logic, schemas, and state transitions, but does *not* guarantee live external third-party API availability.

Provider Subsystem	Mock Tested	Local Real	Live External	Confidence Level
LLM (GroqProvider)	YES	YES	PARTIAL	Moderate (Subject to rate limits)
TTS (EdgeTTSProvider)	YES	YES	YES	High (Free Microsoft endpoint)
Avatar (HeyGenProvider)	YES	NO	NO	Low / Guarded in Mock Mode
Visuals (Pillow)	YES	YES	N/A (Local)	High (Deterministic offline)
Subtitles (Deterministic)	YES	YES	N/A (Local)	High (Zero-drift math)

Composition (FFmpeg)	YES	YES	N/A (Local)	High (When binary is on PATH)

3. Comprehensive Real-World Defect Register

The following issue register catalogs all confirmed and suspected defects identified during testing. Per project constraints, **no functional application changes or bug fixes have been applied**.

[ISSUE-001] Script Generation Pipeline Failure & Diagnostic Swallowing — CRITICAL

Project manifests (e.g. proj_7b26899ede40) fail at Stage 1 with 'Script generation failed: No final script produced', halting downstream stages.
Confirmed Cause 1: orchestrator.py inspects state.get('error_message'), but VideoScriptState defines errors as state.errors, swallowing actual validation issues.
Confirmed Cause 2: Space-separated WPM calculation underestimates Arabic speech duration, triggering false DURATION_MISMATCH rejections on short videos (30s ± 20%) that exhaust single-revision limits.

[ISSUE-002] Avatar Engine Locked in Mock Mode Without UI Control — HIGH

Streamlit displays 'Avatar Engine: Mock Mode Active (0 Credits Used)'. The system generates placeholder colored canvases even when a valid HEYGEN_API_KEY exists.
Confirmed Cause: Hardcoded cost-safety defaults (DEFAULT_AVATAR_PROVIDER=mock, ENABLE_LIVE_API_TESTS=False) prevent live execution. No runtime UI switch exists to toggle live HeyGen generation safely.

[ISSUE-003] Hardcoded LLM Model Label in Streamlit Sidebar — MEDIUM

Streamlit sidebar displays 'LLM Provider: Groq (Llama-3.3-70b)' while backend configuration is GROQ_MODEL=openai/gpt-oss-20b.
Confirmed Cause: Stale static string in frontend/streamlit_app.py line 84 that does not query backend health metadata.

[ISSUE-004] Manifest Diagnostic Error Duplication on Retries — MEDIUM

Project manifests accumulate duplicate identical error messages when pipeline runs are retried.
Confirmed Cause: Unconditional append() in orchestrator.py without stage-level error resetting or deduplication.

[ISSUE-005] Base-Language and Source-Content Mismatch — HIGH

Supplying English source text while selecting base_language=ar-SA and target_languages=[ar-SA] causes the script writer to author Arabic directly, while LocalizationService skips cultural adaptation.
Confirmed Cause: Architectural assumption that base_language matches source_content language. Bypasses cultural adaptation and domain terminology checks.

[ISSUE-006] Language-Agnostic WPM Duration Estimation — MEDIUM

Space-split token formula (len(narration.split()) / 140 * 60) treats morphologically dense Arabic and Bengali identically to English.
Likely Cause: Latin-centric WPM heuristic produces artificially low duration figures for non-English scripts, causing duration rejections.

[ISSUE-007] Bidirectional Arabic Subtitle Rendering in Standard FFmpeg — LOW

Burning Arabic subtitles with standard FFmpeg subtitles filter may render disconnected or reversed characters on Windows builds lacking FriBidi/HarfBuzz font-shaping.
Likely Cause: System FFmpeg binary variation. Recommended direction: Pre-shape Arabic text with python-bidi/arabic-reshaper or embed styled ASS subtitles.

[ISSUE-008] Single-Node Local File Persistence Without Locking — LOW

All project manifests and stage states are stored in un-locked JSON files on local disk (storage/projects//project.json).
Confirmed Cause: Local prototype design. Concurrency risks exist under high asynchronous load without an ACID database.

[HIST-001] Frontend/Backend HTTP 422 Contract Mismatch — RESOLVED

Streamlit previously returned HTTP 422 Unprocessable Content due to flat payload structure and mismatched field names. **Resolution:** Implemented canonical serialization in serialize_create_project_payload(), added APIClientError handling, and protected with regression tests in test_frontend.py and test_api_projects.py. Verified returning HTTP 201 Created on live server.

4. Security & Secret Audit Verification

A full recursive audit of all tracked and untracked files confirmed that **zero API keys, credentials, or private tokens are present in source code, documentation, or commit staging**. The real .env file is strictly ignored by .gitignore, and only sanitized placeholders exist in .env.example. All runtime project media and virtual environments are excluded.

5. Platform Limitations & Future Roadmap

Current Architectural Limitations:

- Pre-TTS duration estimation cannot perfectly predict synthetic voice inflection or pauses.
- Script generation depends entirely on third-party Groq API availability and rate limits.
- Real avatar synthesis requires paid HeyGen credits; testing remains guarded in mock mode by default.
- Native composition requires host-installed FFmpeg binaries; cloud deployment will require containerization.
- Video composition runs synchronously in-process rather than via an asynchronous worker queue (Celery/Redis).

Prioritized Future Roadmap:

Priority 1 (Pre-Demo): Synchronize orchestrator error logging with VideoScriptState.errors, add locale-aware WPM multipliers, connect Streamlit badges dynamically to /api/v1/health.

Priority 2 (Pre-Production): Migrate persistence to PostgreSQL, offload FFmpeg rendering to Celery/Redis, pre-shape Arabic subtitle text with python-bidi, add user authentication.

Priority 3 (Enhancements): Expand language library (Spanish, French, German), add karaoke subtitle highlight effects via Whisper, implement WebSocket progress streaming.

AUDIT CONCLUSION: The platform architecture demonstrates robust separation of concerns, complete schema contracts, and 100% automated regression test coverage across all internal components. Real live execution is constrained by the documented duration validation heuristics and intentional mock avatar safety guards.

STATUS: AUDIT COMPLETE — NO BUG FIXES APPLIED